

Multi-Turn Context Budget, Worked Out by Hand

A long-running agent accumulates tokens every turn until it must summarise. We derive when the 70% trigger fires, what summarisation reclaims, and the sawtooth it produces.

What this document does. A multi-turn MCP agent grows its message history every turn; left unchecked it overflows the context window. The standard fix is: when token count crosses $\sim 70\%$ of the window, summarise the old turns. This document derives how many turns that buys, why the threshold is below 100%, what the compression reclaims, and the steady-state sawtooth of a long session.

Concepts covered: linear token accumulation · the 70% trigger · headroom for reply + next tool result · compression ratio · the summarise sawtooth · steady-state cadence.

Internal learning material. Numbers reproducible from the appendix code. v1.0

1 The setup: a window and a per-turn cost

window $W = 200,000$ tokens, summarise trigger $\theta = 0.70$, per turn $t = 3,000$ tokens.

Each turn (user message + tool call + tool result + assistant reply) adds about $t = 3,000$ tokens to the running history. The agent summarises once the history crosses $\theta W = 0.70 \times 200,000 = 140,000$ tokens.

Intuition

History is append-only: every turn the model must re-read everything before it. Cost grows linearly with turns until you compress. The trigger is a high-water mark: cross it, replace the old transcript with a short summary, and keep going. The only questions are *when* it fires and *how much* compressing buys back.

2 Step 1 — When the trigger fires

After n turns the history holds $S(n) = n \cdot t$ tokens (plus a fixed system preamble we fold into t). The trigger fires at the first n with $S(n) \geq \theta W$:

$$n^* = \left\lceil \frac{\theta W}{t} \right\rceil = \left\lceil \frac{140,000}{3,000} \right\rceil = \lceil 46.67 \rceil = \boxed{47 \text{ turns}}.$$

So the agent runs 47 turns before its first summarisation.

3 Step 2 — Why 70%, not 100%

The threshold sits below the window because the window must also hold what hasn't been counted yet: the model's *reply* and the *next tool result*, both of which arrive *after* the trigger check. Reserve

$$\text{headroom} = (1 - \theta) W = 0.30 \times 200,000 = 60,000 \text{ tokens}$$

for reply + next result + safety margin. Trigger at 100% and the very next generation has nowhere to go — truncated output or a hard error.

Watch out

The right θ depends on how big a single turn can be. If one tool can return a 50,000-token result, a 30,000-token headroom ($\theta = 0.85$) is already too tight — the result alone overflows. Set $1 - \theta$ to at least your worst-case single-turn size, not a round number.

4 Step 3 — What summarisation reclaims

Summarisation compresses the old history of H tokens into a summary of rH , where r is the compression ratio (say $r = 0.1$, a $10\times$ squeeze). At the trigger, $H \approx 140,000$, so:

$$\text{after} = rH = 0.1 \times 140,000 = 14,000 \text{ tokens}, \quad \text{reclaimed} = (1 - r)H = 126,000 \text{ tokens}.$$

The history drops from 140k back to 14k — 90% of the budget freed in one step.

5 Step 4 — The steady-state sawtooth

After the first summary the floor is no longer 0 but 14,000 (the retained summary). Each subsequent cycle climbs from that floor to the 140,000 trigger, so the number of turns *between* summarisations is

$$n_{\text{cycle}} = \left\lceil \frac{\theta W - rH}{t} \right\rceil = \left\lceil \frac{140,000 - 14,000}{3,000} \right\rceil = \left\lceil \frac{126,000}{3,000} \right\rceil = 42 \text{ turns}.$$

The token count traces a sawtooth: rise 14k $\rightarrow$ 140k over 42 turns, drop back to 14k, repeat — indefinitely, so the agent never overflows.

phase	turns	tokens span
first fill	47	0 $\rightarrow$ 140,000
summarise	—	140,000 $\rightarrow$ 14,000
each cycle after	42	14,000 $\rightarrow$ 140,000

Intuition

The retained summary is permanent overhead: every cycle starts 14k in the hole, so steady-state cycles (42 turns) are slightly shorter than the first (47). A smaller r (tighter summary) lengthens cycles but risks dropping detail; a larger r keeps more context but summarises more often. r trades fidelity against cadence.

6 The argument, at a glance

#	Idea	Formula	Value
1	Trigger threshold	θW	140,000
1	Turns to first trigger	$\lceil \theta W / t \rceil$	47
2	Reserved headroom	$(1 - \theta)W$	60,000
3	After summary	rH	14,000
3	Reclaimed	$(1 - r)H$	126,000
4	Turns per cycle	$\lceil (\theta W - rH) / t \rceil$	42

7 Equation cheat-sheet

History after n turns	$S(n) = n t$
First-trigger turn	$n^* = \lceil \theta W / t \rceil$
Reserved headroom	$(1 - \theta) W$
Post-summary size	rH (ratio r)
Reclaimed tokens	$(1 - r)H$
Turns per steady cycle	$\lceil (\theta W - rH) / t \rceil$

A Reference implementation

```
import math
W, theta, t, r = 200_000, 0.70, 3_000, 0.10

trigger = theta * W                # 140000
n_first = math.ceil(trigger / t)   # 47
H = trigger
after = r * H                      # 14000
reclaimed = (1 - r) * H            # 126000
n_cycle = math.ceil((trigger - after) / t) # 42
print(n_first, after, reclaimed, n_cycle)  # 47 14000.0 126000.0 42
```

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